Exercise Week 3 Solutions

Advanced Applied Statistics

Fall 2025

Problem 1*

$$f_X(x) = \lambda e^{-\lambda x}, \qquad x \geq 0,\ \lambda > 0.$$

$$\int_0^\infty \lambda e^{-\lambda x}\, dx = \lim_{b\to\infty} \big[-e^{-\lambda x} \big]_{x=0}^{x=b} = \lim_{b\to\infty} \big(-e^{-\lambda b} + e^0 \big) = 0 + 1 = 1.$$

Problem 2*

(a)

$$L(\theta) = \theta^x (1-\theta)^{1-x}$$

(b) The log-likelihood $\ell = \log L(\theta)$. So,

$$\begin{aligned}
\ell(\theta) &= x\log(\theta) + (1-x)\log(1-\theta) \\
&= x\log(\theta) + \log(1-\theta) - x\log(1-\theta) \\
&= \log(1-\theta) + x\log\left(\frac{\theta}{1-\theta}\right)
\end{aligned}$$

(c) The score function $\dot{\ell}(\theta)$ is the first derivative of the score function. So,

$$\begin{aligned}
\dot{\ell}(\theta) &= -\frac{1}{1-\theta} + x\left(\frac{1-\theta}{\theta}\right)\frac{(1-\theta)+\theta}{(1-\theta)^2} \\
&= -\frac{1}{1-\theta} + x\left(\frac{1}{\theta}\right)\frac{1}{1-\theta} \\
&= \frac{x-\theta}{\theta(1-\theta)}
\end{aligned}$$

(d) Lastly, we calculate the Fisher Information by taking the variance of the score function:

$$\begin{aligned}
I(\theta) &= V\left(\frac{X-\theta}{\theta(1-\theta)}\right) = E\left(\frac{(X-\theta)^2}{\theta^2(1-\theta)^2}\right) \\
&= \frac{E[(X-\theta)^2]}{\theta^2(1-\theta)^2} = \frac{V[X]}{\theta^2(1-\theta)^2} \\
&= \frac{\theta(1-\theta)}{\theta^2(1-\theta)^2} = \frac{1}{\theta(1-\theta)}
\end{aligned}$$

Problem 3

(a)

If we only care about the first k features, then we can use Schurr's Complement to model the conditional distribution of those first k features (see problem 4).

(b)

In high-dimensional settings (think datasets of genomes, or pixel data in images), inverting the full covariance is costly. Using the Schurr complement, we can compute the conditional covariance for a smaller subset of features efficiently.

Problem 4

If the joint pdf of X and Y is given by

$$\pi(x, y) \propto \exp\left(-\frac{1}{2}\begin{bmatrix} x - x_0 \\ y - y_0 \end{bmatrix}^{\mathrm{T}} \begin{bmatrix} \Gamma_{11} & \Gamma_{12} \\ \Gamma_{21} & \Gamma_{22} \end{bmatrix}^{-1} \begin{bmatrix} x - x_0 \\ y - y_0 \end{bmatrix}\right)$$

then

$$\pi(y|x) \propto \exp\left(-\frac{1}{2}\left[y - \bar{y}\right]^{\mathrm{T}} \tilde{\Gamma}_{11}^{-1}\left[y - \bar{y}\right]\right)$$

where $\tilde{\Gamma}_{11}$ is the Schurr complement and $\bar{y} = y_0 + \Gamma_{21}\Gamma_{11}^{-1}(x - x_0)$.

Problem 5

Recall that the pdf of the exponential distribution is given by:

$$f(x; \lambda) = \lambda e^{-\lambda x}, \qquad x \geq 0,\ \lambda > 0.$$

and we would like to represent it in the form:

$$f(x; \eta) = h(x)g(\theta)\exp\big(\eta(\theta) \cdot T(x)\big)$$

Thus, $\eta(\theta) = -\lambda$, $h(x) = \mathbf{1}_{\{x \geq 0\}}$, $T(x) = x$, and $g(\theta) = \lambda$.